# Matteo Giorgetti

Computer Science & Artificial Intelligence | Software Engineering

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Computer Science and Artificial Intelligence student at IE University, building native applications, reusable simulation tools and robotics software. Research experience in AI and robotics, with projects spanning backend systems, computer vision and on-device AI.

## EDUCATION

**BSc Computer Science and Artificial Intelligence** 2024 - Expected May 2028
IE University | Madrid, Spain

**International Baccalaureate Diploma** Graduated May 2024
H-FARM International School | Roncade, Italy | 40/45

## EXPERIENCE

**Research Assistant** 2024 - Present
IE University AI & Robotics Lab | Madrid, Spain

- Developed software for robotics research and demonstrations, collaborating with researchers on integration and troubleshooting.
- Worked on MOSAIC, combining robot fleet coordination with calibrated floor visualization, recording and observation-only replay; movement authority remains in the robot application.

**Donation Automation Developer** Mar 2023 - Feb 2025
Fondazione Dottor Sorriso | Milan, Italy

- Built Python automation to extract donor names from 1,000+ inconsistent donation records and enter donations in the foundation's existing platform.
- Combined model-assisted extraction, caching and Selenium automation, reducing manual entry by 90% and saving an estimated 40 hours of administrative work per month.

## SELECTED ENGINEERING PROJECTS

**Noty** 2026
Swift · SwiftUI · AppKit · SQLite

- Developed a native macOS note application using Swift, SwiftUI and AppKit, with SQLite storage and notes attached to application windows.
- Separated focus, window positioning and visibility, with recoverable editing and explicit privacy controls for optional AI access.

**PaddockJS** 2026
JavaScript · PixiJS

- Built a reusable JavaScript/PixiJS racing simulator with composable browser views and a shared headless reset/step API.
- Separated grid browsing from simulation state and exposed configurable rules and policy-evaluation interfaces. The package powers the project race on my portfolio.

## SELECTED ENGINEERING PROJECTS / CONTINUED

### Dream2Detect
2026

Python · PyTorch · Computer vision

- Designed and implemented the end-to-end pipeline for a group project: synthetic package images, human label review, model training and evaluation on real photographs.
- The reported ensemble achieved 64.16% within one severity band on 385 real photos; exact-band accuracy was 21.04%.

### Who's On Repeat
2026

Python · SQLite · JavaScript

- Designed and implemented a multiplayer music game as a solo course project, with a Python backend, SQLite transactions and server-timed scoring.
- Separated room and game rules, protected answers until reveal, and integrated Spotify alongside an offline demo.

### VIGIL
2026

Python · FastAPI · Paramiko · ROS

- Developed a robot-fleet operations dashboard using FastAPI, Paramiko and ROS interfaces, with browser-based SSH terminals.
- Defined reusable diagnostic and recovery workflows, WebSocket terminal streaming and per-robot coordination to prevent conflicting jobs.

## COMMUNITY & LEADERSHIP

**OpenAI Codex Ambassador** Feb 2026 - Present

Explore agentic coding workflows and share practical approaches to AI-assisted software development.

**Google Developer Group / Students Club** 2024 - Present

Progressed from Tech Guru to club lead, contributing to technical workshops and community projects.

### Bot Battle · Game Environment & Judging
Mar - Apr 2025

Created the modular game environment, helped coordinate bot simulations and judged final presentations in a two-week AI hackathon with seven presenting teams.

### Tech Roulette · Hackathon Judge
2026

Served on the judging panel for the student hackathon's final project presentations.

## TECHNICAL SKILLS

**Programming:** Python, Swift, JavaScript, SQL, Java, C++.

**Native applications:** SwiftUI, AppKit, SQLite, macOS Accessibility, Android.

**Backend & systems:** FastAPI, Django, PostgreSQL, HTTP / WebSockets, TCP / UDP, Docker, Git, Automated testing.

**AI & machine learning:** PyTorch, OpenCV, NumPy, Computer vision, Synthetic data, MLX, LLM integration.

**Robotics:** ROS 2, OptiTrack, Robot perception, Telemetry, Calibration.

**Courses:** LangChain Academy: Foundation: Introduction to LangChain - Python (Apr 2026); Anthropic: Introduction to Agent Skills (Mar 2026); Anthropic: Claude Code in Action (Mar 2026); Amazon Web Services: AWS Educate Getting Started with Compute (Oct 2025).

**Languages:** Italian - native; English - native; Spanish - intermediate; French - basic.